

Cybersecurity Lab Setup for Penetration Testing (Beginner)

Full Setup Documentation



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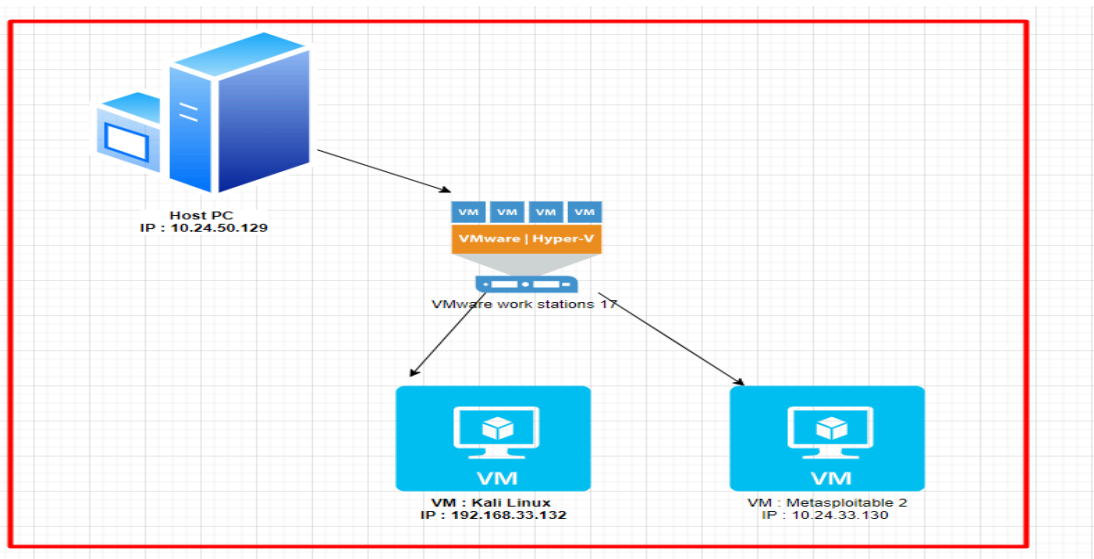
1. Introduction

This lab is designed for beginners in penetration testing. You will set up a virtual environment to safely practice attacking vulnerable systems. We will use **Kali Linux**, a popular OS for ethical hacking, and **Metasploit**, a framework for developing and executing exploits.

2. Lab Requirements

- **Hardware:**
 - A PC with at least 8 GB RAM (16 GB recommended) and 40 GB of free disk space.
- **Software:**
 - Virtualization software (VMware Workstation/Player or VirtualBox)
 - Kali Linux ISO file
 - Metasploitable2 VM (target machine)

Lab Design :



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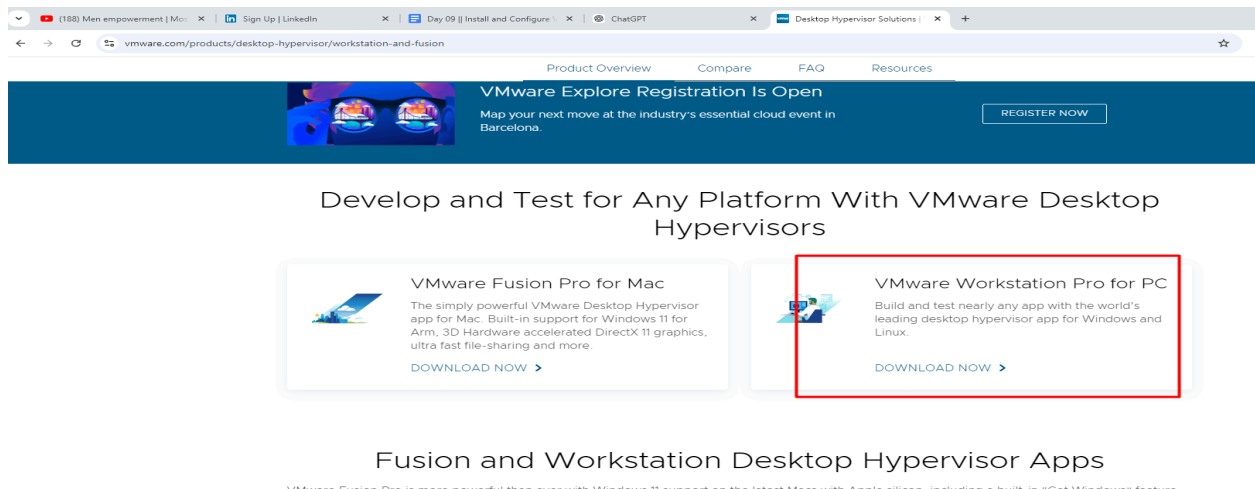
3. Installing Virtualization Software (VMware/VirtualBox)

To set up a lab, you will need to run virtual machines (VMs). Download and install a virtualization platform like **VMware Workstation/Player** or **VirtualBox**.

Steps for VMware Player:

1. Download VMware Workstation Player from the official site.

Link : <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>



2. Follow the installation prompts.
3. Once installed, launch VMware Player.

Resource : <https://youtu.be/zCanGptZOTE?si=jW2ZdhvvS8ljGvIJ>

General Host OS Requirements

VMware Workstation Pro and Player run on most 64-bit Windows or Linux host operating systems:

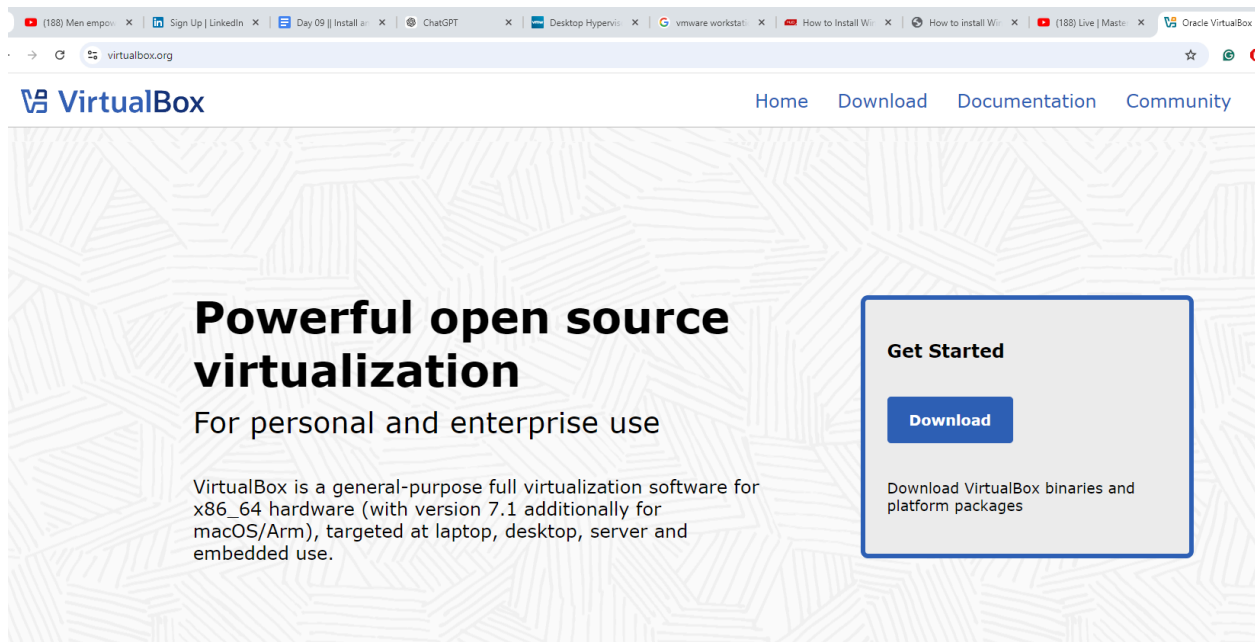
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- Windows 10
- Windows Server 2019
- Windows Server 2016
- Windows Server 2012
- Windows 8
- Ubuntu
- Red Hat Enterprise Linux
- CentOS
- Oracle Linux
- openSUSE
- SUSE Linux Enterprise Server

Steps for VirtualBox:

1. Download VirtualBox from its official website.

<https://www.virtualbox.org/>



2. Follow the installation steps.
3. Launch VirtualBox after installation.

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Creating a beginner-friendly cybersecurity lab with **Kali Linux** and **Metasploit** is a great way to start learning penetration testing. Here's a step-by-step guide and documentation to help you set up your lab:

Video Resource : https://youtu.be/JL5JMzKYPE0?si=p4CwFx631sOO0_78

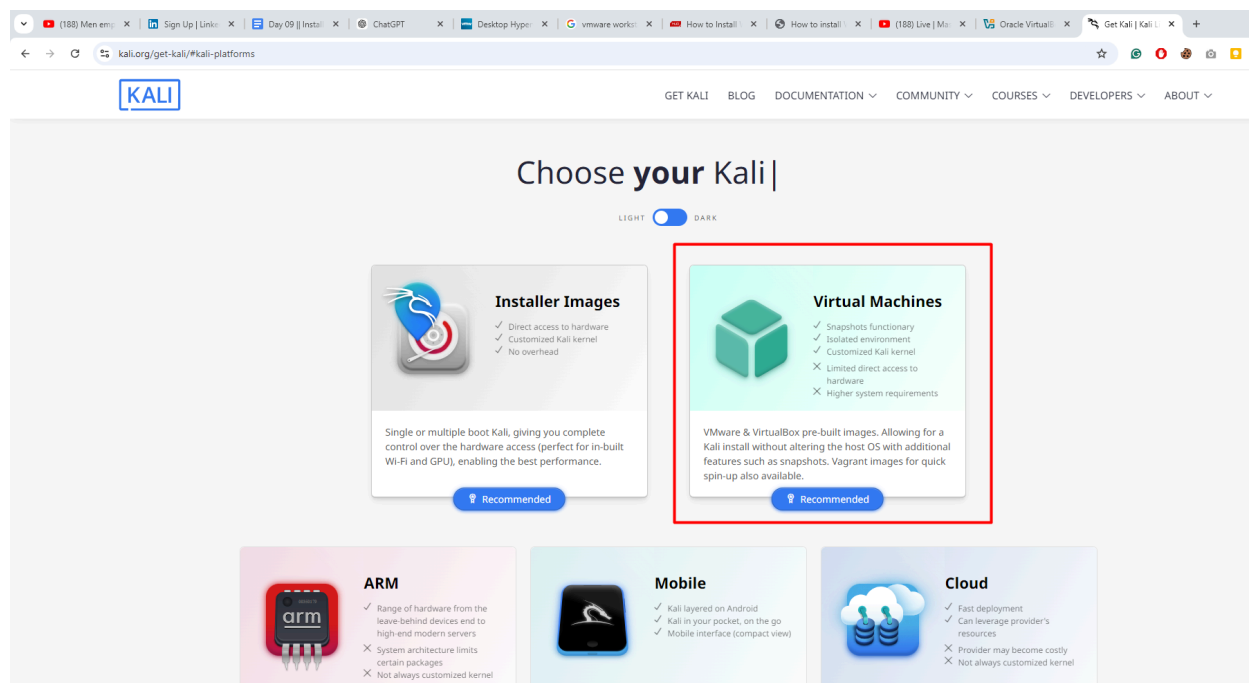
4. Installing Kali Linux

Kali Linux is a Linux distribution focused on penetration testing.

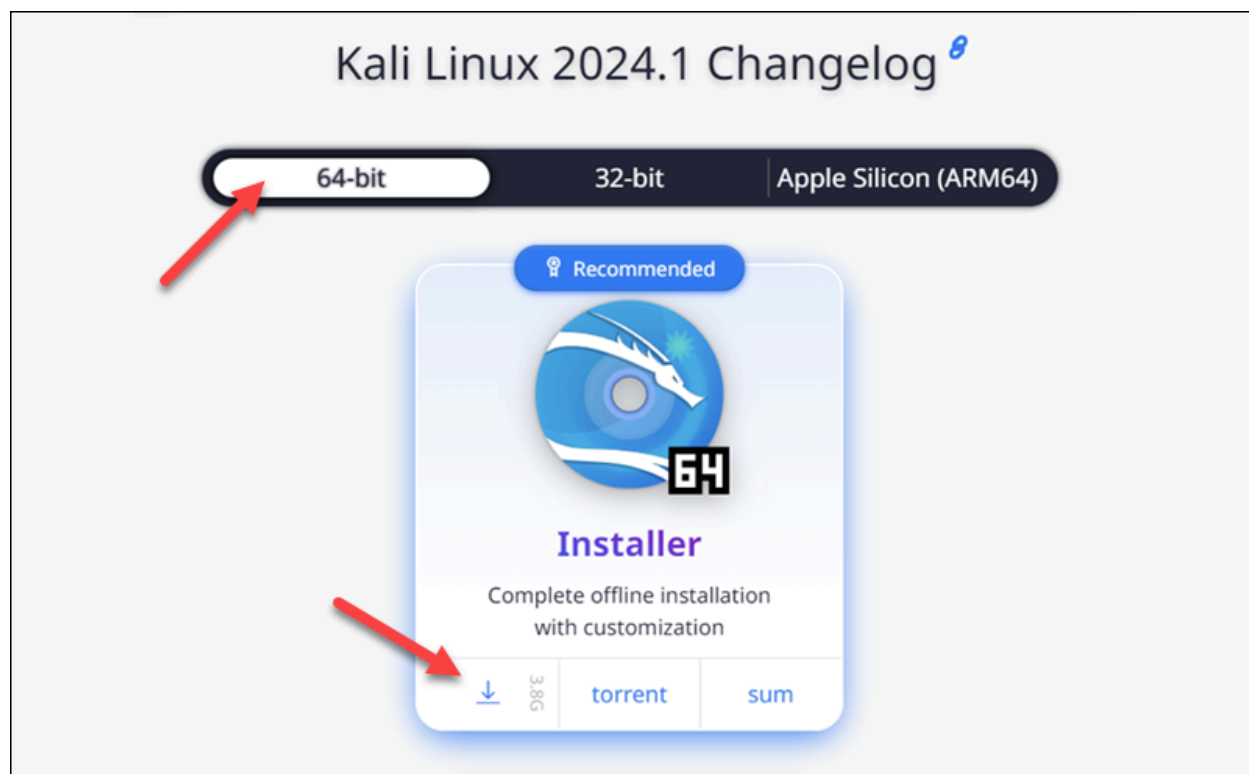
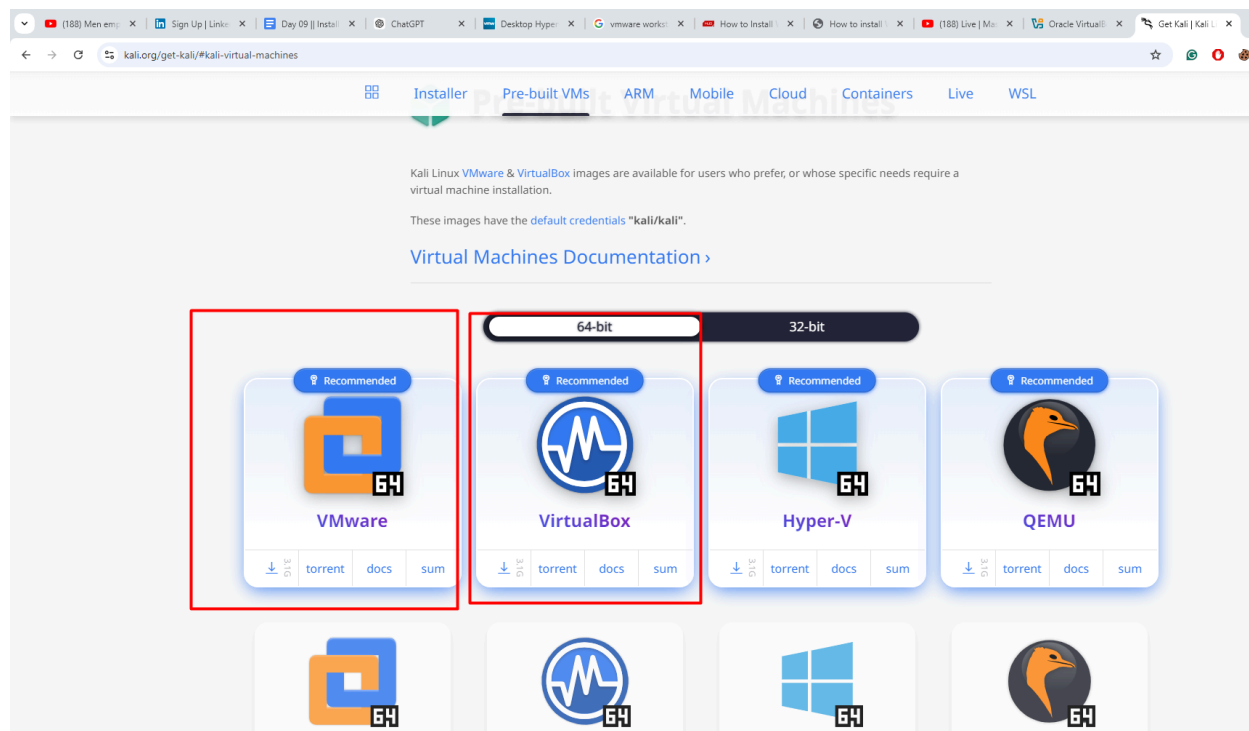
Steps:

1. Download the **Kali Linux ISO** from the official site here.

<https://www.kali.org/get-kali/#kali-platforms>



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2. Open your virtualization software (VMware/VirtualBox).
3. Create a new virtual machine:
 - In VMware: Click **Create a New Virtual Machine**, select the Kali Linux ISO, and follow the installation prompts.
 - In VirtualBox: Click **New**, set the OS type to **Linux**, and point it to the Kali Linux ISO.
4. Configure your VM settings (minimum 2 GB RAM, 20 GB disk space).
5. Follow the installation process (select **Graphical Install**).
6. Once installed, update the OS with



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Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: ✓

Folder:

ISO Image: ✓

Edition:

Type: 64

Version:

☐ Skip Unattended Installation

ⓘ Detected OS type: Ubuntu (64-bit). This OS type cannot be installed unattendedly. The install needs to be started manually.

Help Expert Mode Back **Next** Cancel

Create Virtual Machine

Hardware

You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count. Enabling EFI is also possible.

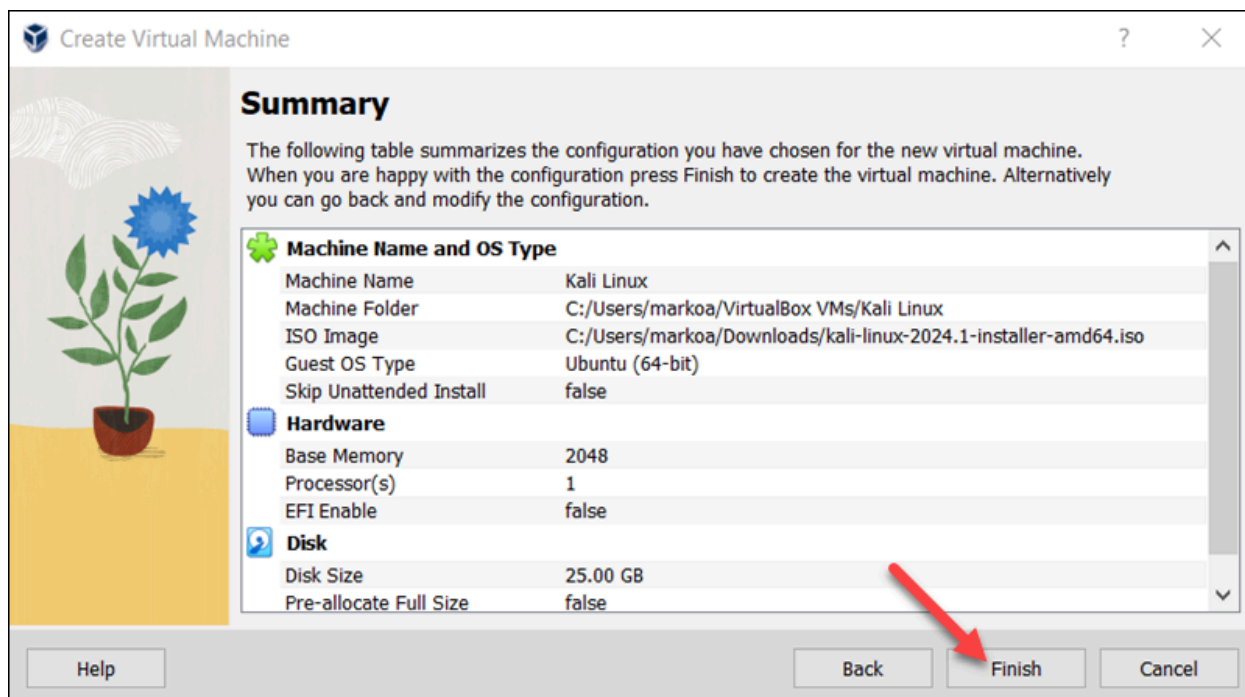
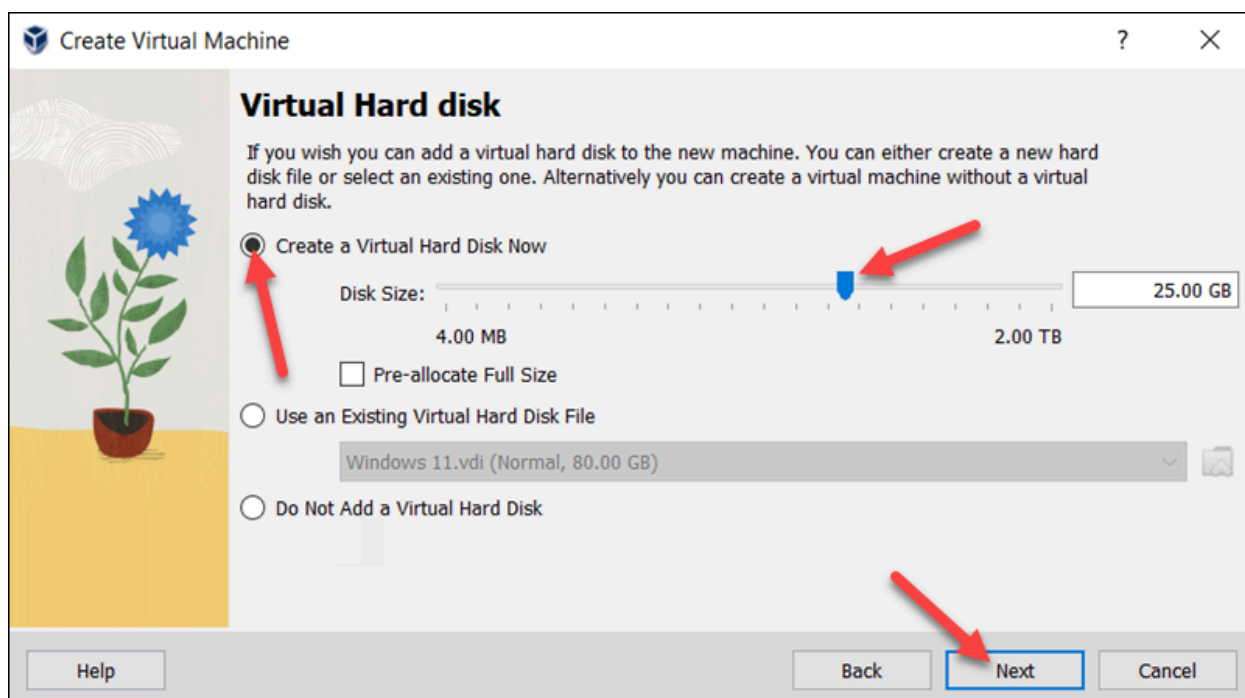
Base Memory: 2048 MB

Processors: 1

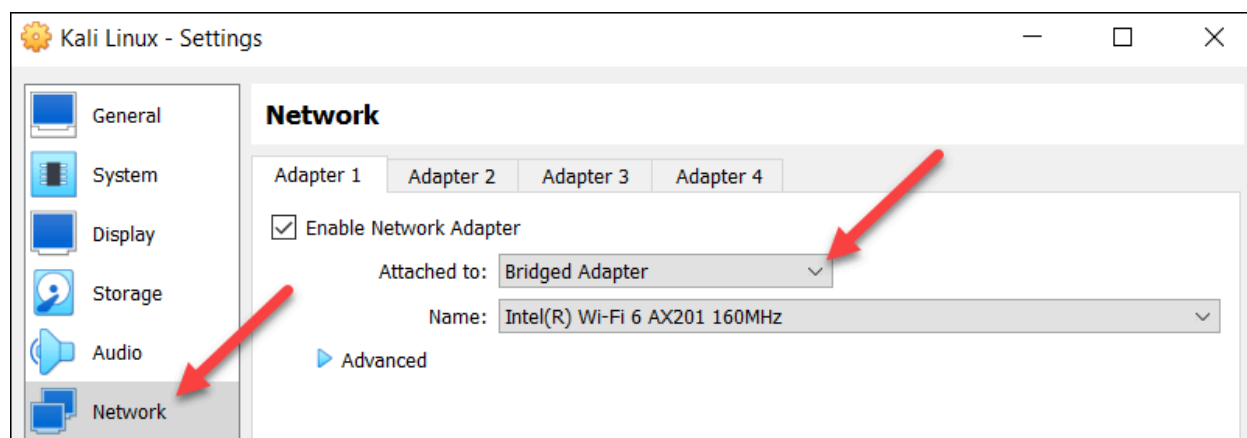
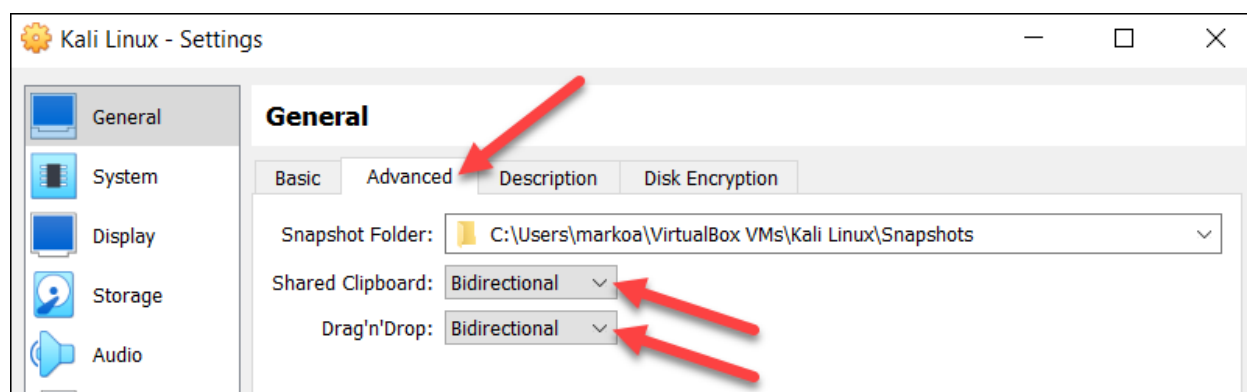
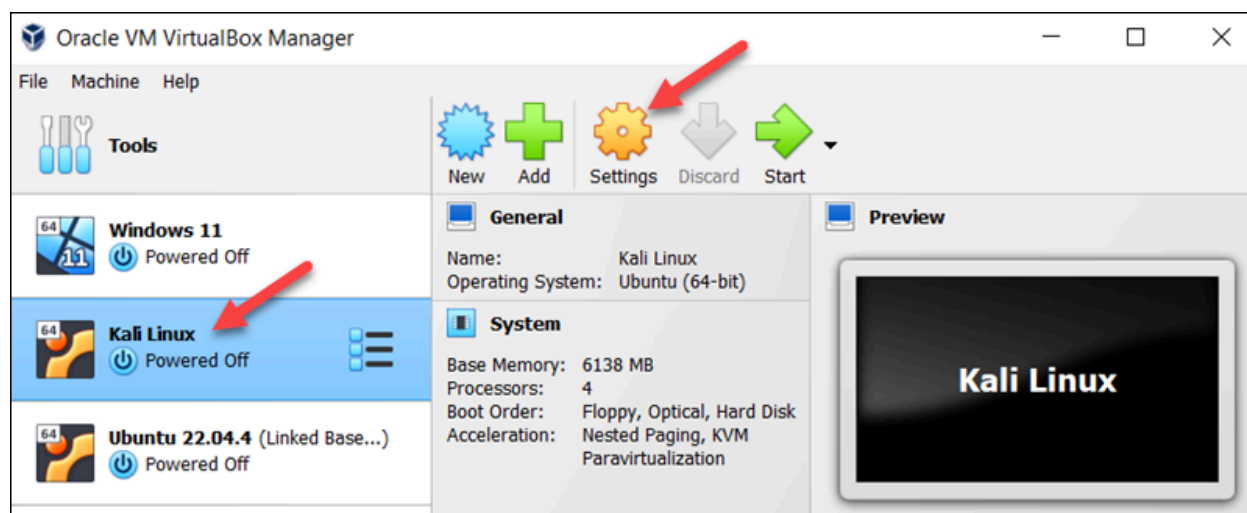
☐ Enable EFI (special OSes only)

Help Back **Next** Cancel

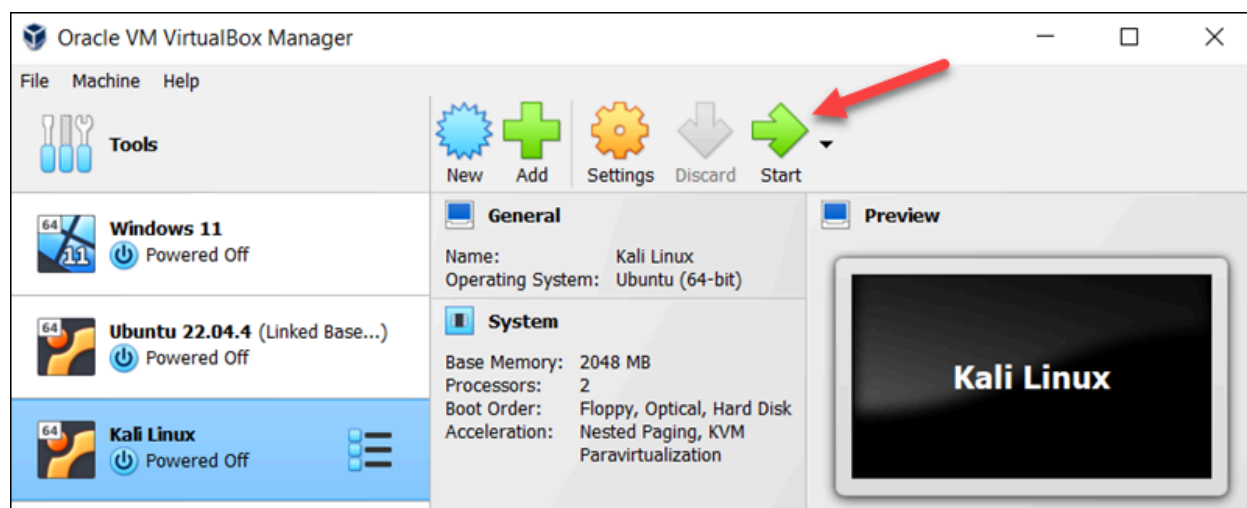
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Cybersecurity Lab Setup for Penetration Testing (Beginner)



5. Setting Up a Target Machine (Metasploitable2)

Metasploitable2 is a deliberately vulnerable Linux machine. It will be your target for penetration testing.

Steps:

1. Download **Metasploitable2** from [here](#).

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The screenshot shows the SourceForge project page for Metasploitable. The browser's address bar displays `sourceforge.net/projects/metasploitable/`. The SourceForge header includes navigation links for Open Source Software, Business Software, SourceForge Podcast, and Resources, along with a search bar. The project page for Metasploitable features a dark theme with the project name in large white text. Below the name, it states "Metasploitable is an intentionally vulnerable Linux virtual machine" and "Brought to you by: rapid7user". The page shows a 4.5-star rating from 10 reviews, 15,906 downloads this week, and a last update date of 2019-08-19. A prominent green "Download" button is visible, along with "Get Updates" and "Share This" buttons. A tabbed interface at the bottom shows the "Summary" tab selected, displaying a description of the VM, its default login/password, and contact information. The "Categories" section lists "Security, Penetration Testing, VMware" and the "License" section lists "BSD License, GNU General Public License version 2.0 (GPLv2)". On the right side, there are sections for "Recommended Projects" (including OWASP Broken Web Applications Project and Virtual Hacking Lab) and "Top Searches" (including vulnerable machine, metasploitable-2, metasploitable, and metasploitable 3 virtual machine).

sourceforge.net/projects/metasploitable/

SOURCEFORGE

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Metasploitable

Metasploitable is an intentionally vulnerable Linux virtual machine
Brought to you by: [rapid7user](#)

★★★★★ 10 Reviews Downloads: 15,906 This Week Last Update: 2019-08-19

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[Summary](#) [Files](#) [Reviews](#) [Support](#)

This is Metasploitable2 (Linux)

Metasploitable is an intentionally vulnerable Linux virtual machine. This VM can be used to conduct security training, test security tools, and practice common penetration testing techniques.

The default login and password is msfadmin:msfadmin.

Never expose this VM to an untrusted network (use NAT or Host-only mode if you have any questions what that means).

To contact the developers, please send email to msfdev@metasploit.com

Categories
[Security, Penetration Testing, VMware](#)

License
BSD License, GNU General Public License version 2.0 (GPLv2)

Recommended Projects

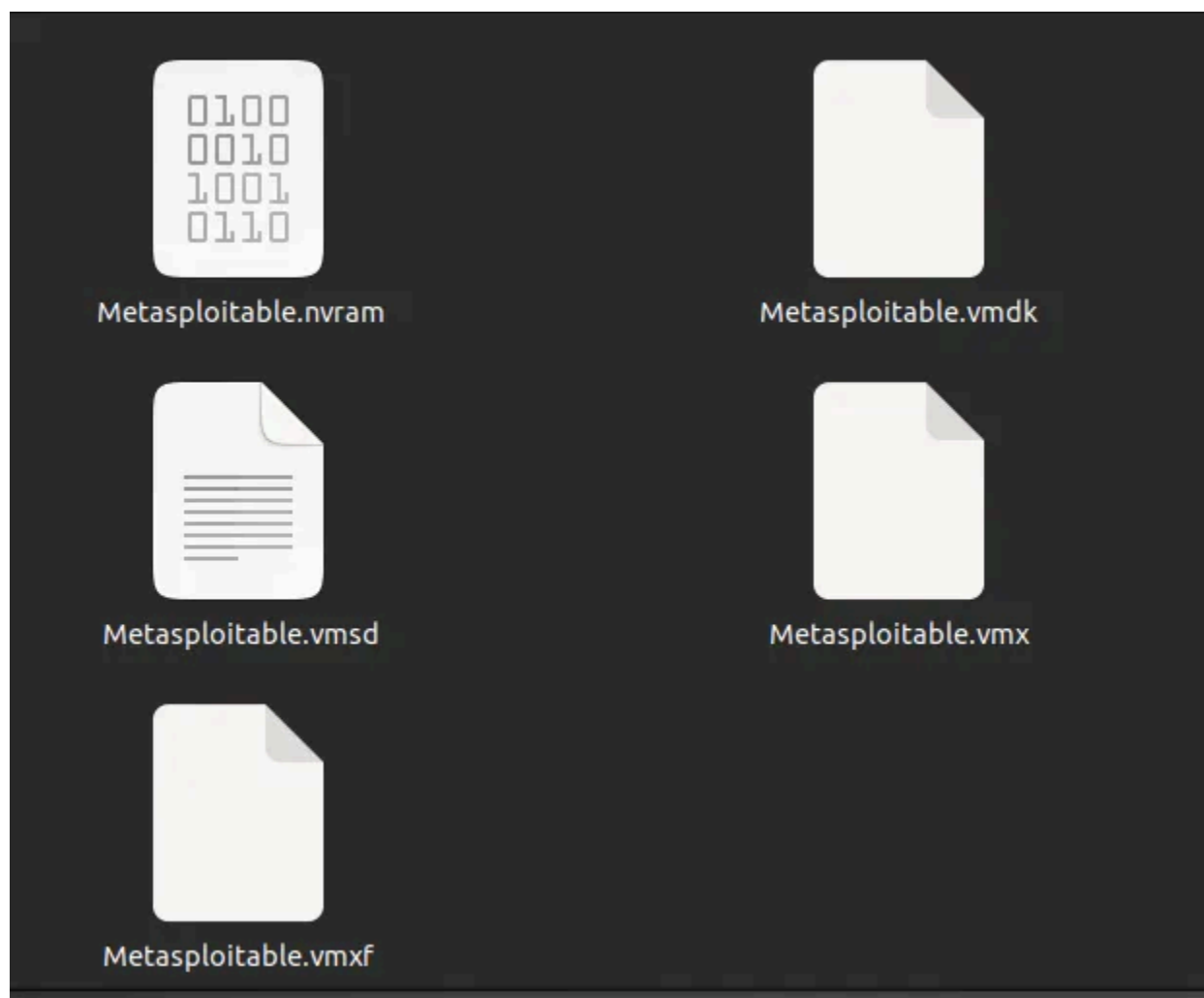
- [OWASP Broken Web Applications Project](#)
Open Web Application Security Project (OWASP)...
- [Virtual Hacking Lab](#)
A mirror of deliberately insecure applications and old...
- [Brakeman](#)
A static analysis security vulnerability scanner for Rub...

Top Searches

- [vulnerable machine](#)
- [metasploitable-2](#)
- [metasploitable](#)
- [metasploitable 3 virtual machine](#)

<https://sourceforge.net/projects/metasploitable/>

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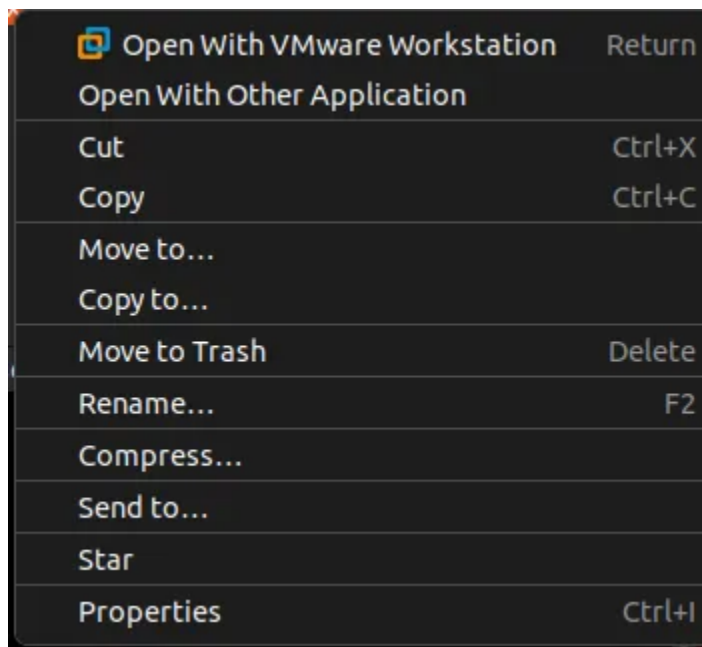


Unzip the downloaded zip file and you should see a new **Metasploitable2-Linux** directory created with the following contents.

Select and right click on the **Metasploitable.vmx** file and you should see the option **Open with VMWare Workstation**. In case you don't see that option, simply click on **Open with Other Application** and select **VMWare** from the list of applications.

2. Open VMware or VirtualBox and import Metasploitable2 as a new VM.

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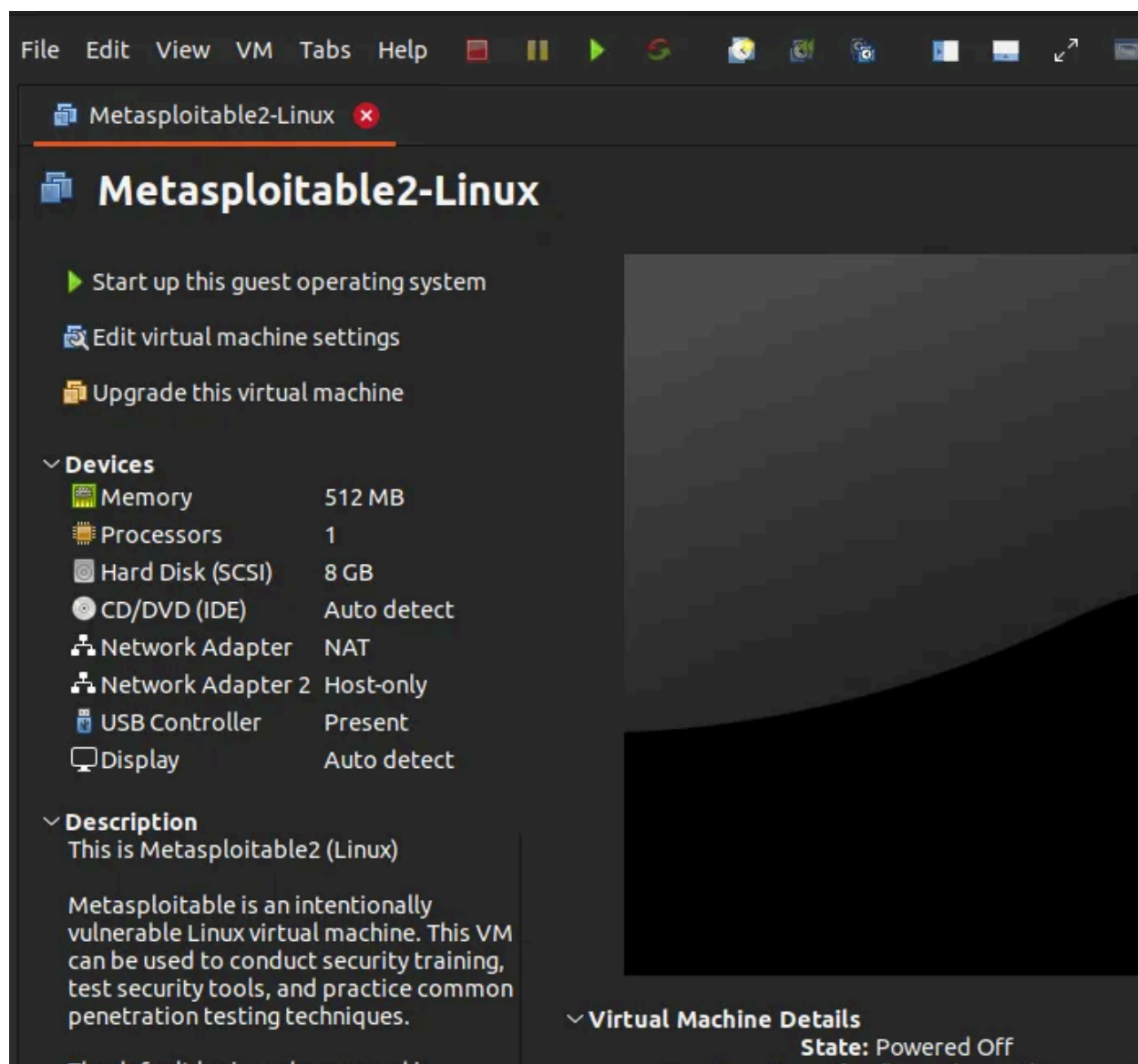


Select and right click on the **Metasploitable.vmx** file and you should see the option **Open with VMWare Workstation**. In case you don't see that option, simply click on **Open with Other Application** and select **VMWare** from the list of applications.

3. Configure it similarly to Kali Linux (2 GB RAM, 10 GB disk space).

After opening the file using VMWare Workstation, you should be able to see the following type of screen.

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4. Boot up the VM. No additional setup is required as this is a vulnerable machine by design.
5. Check the IP address of the Metasploitable2 machine using

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```
Metasploitable2-Linux x

* Starting deferred execution scheduler atd [ OK ]
* Starting periodic command scheduler crond [ OK ]
* Starting Tomcat servlet engine tomcat5.5 [ OK ]
* Starting web server apache2 [ OK ]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to 'nohup.out'
nohup: appending output to 'nohup.out' [ OK ]

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: _
```

The login credentials are already displayed on the terminal, so just login using the default credentials (**msfadmin/msfadmin**)

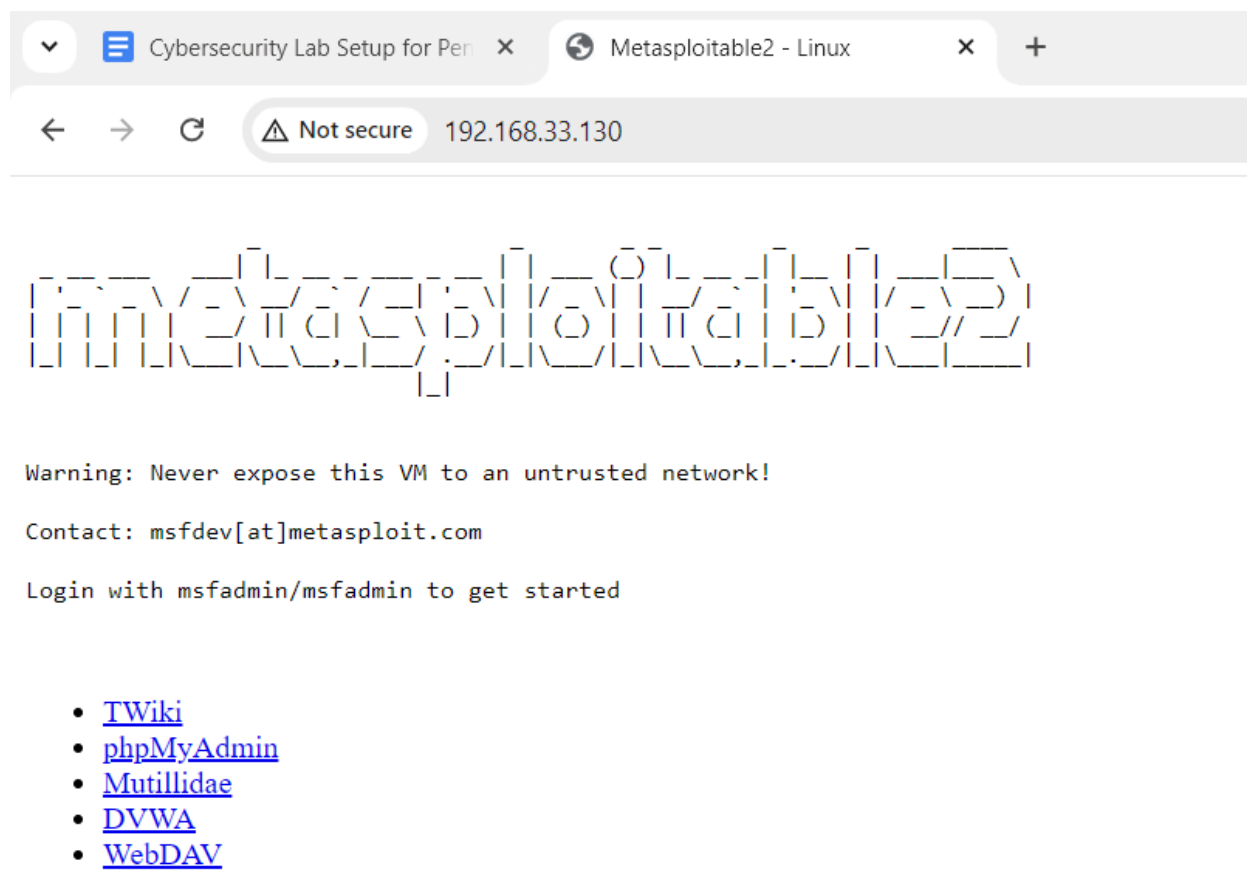
```
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet HWaddr 08:0c:29:0e:73:e6
          inet addr:192.168.33.130 Bcast:192.168.33.255 Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe73:e664 Scope:Link
          UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4631 (4.5 KB) TX bytes:6880 (6.7 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING MTU:16436 Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB) TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

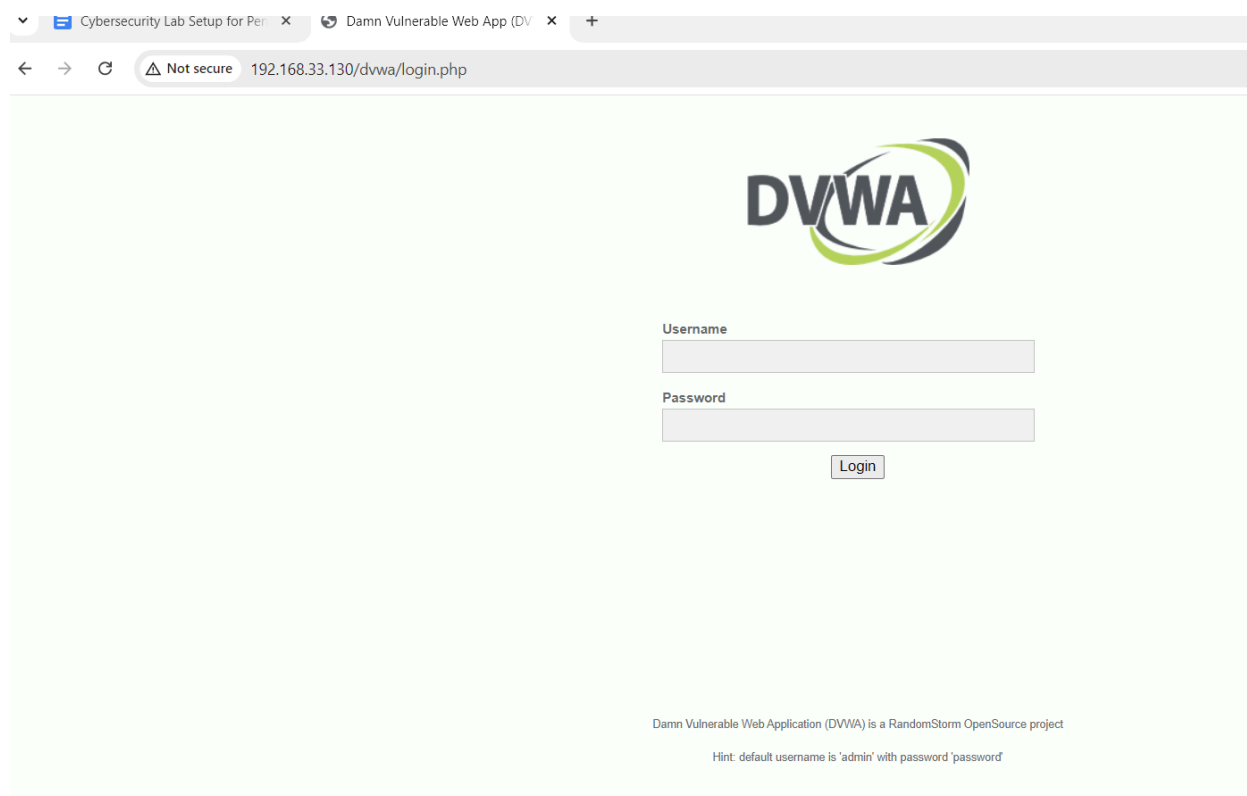
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Launch a web browser on your host machine and enter the IP address (192.168.33.130) assigned to the Metasploitable2 virtual machine. This action will bring up the default Metasploitable2 webpage/homepage.



Click on DVWA to launch the web application. Login using the default credentials- admin/password

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6. Installing and Using Metasploit Framework kali (pre installation in kali)

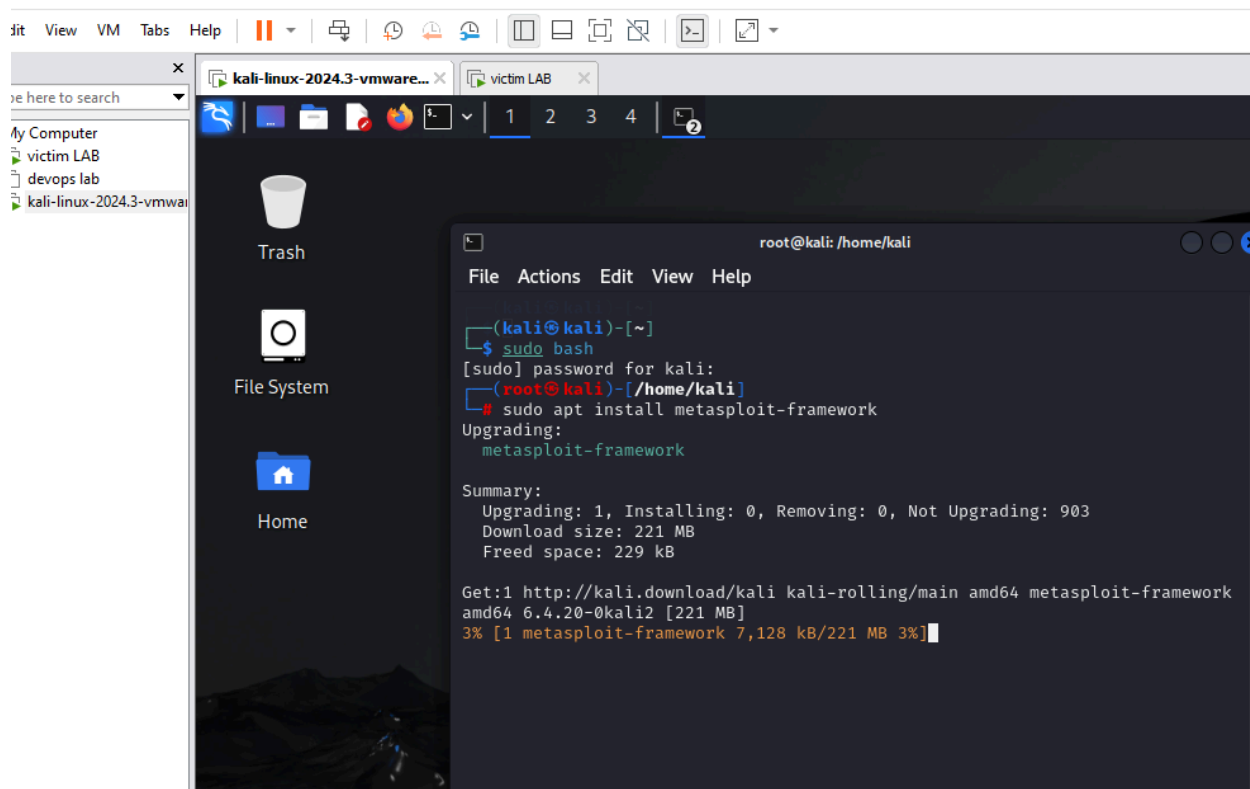
Metasploit is a tool used for developing and executing exploits on vulnerable systems.

Steps:

Metasploit is pre-installed on Kali Linux, but you can update it with:

```
sudo apt install metasploit-framework
```

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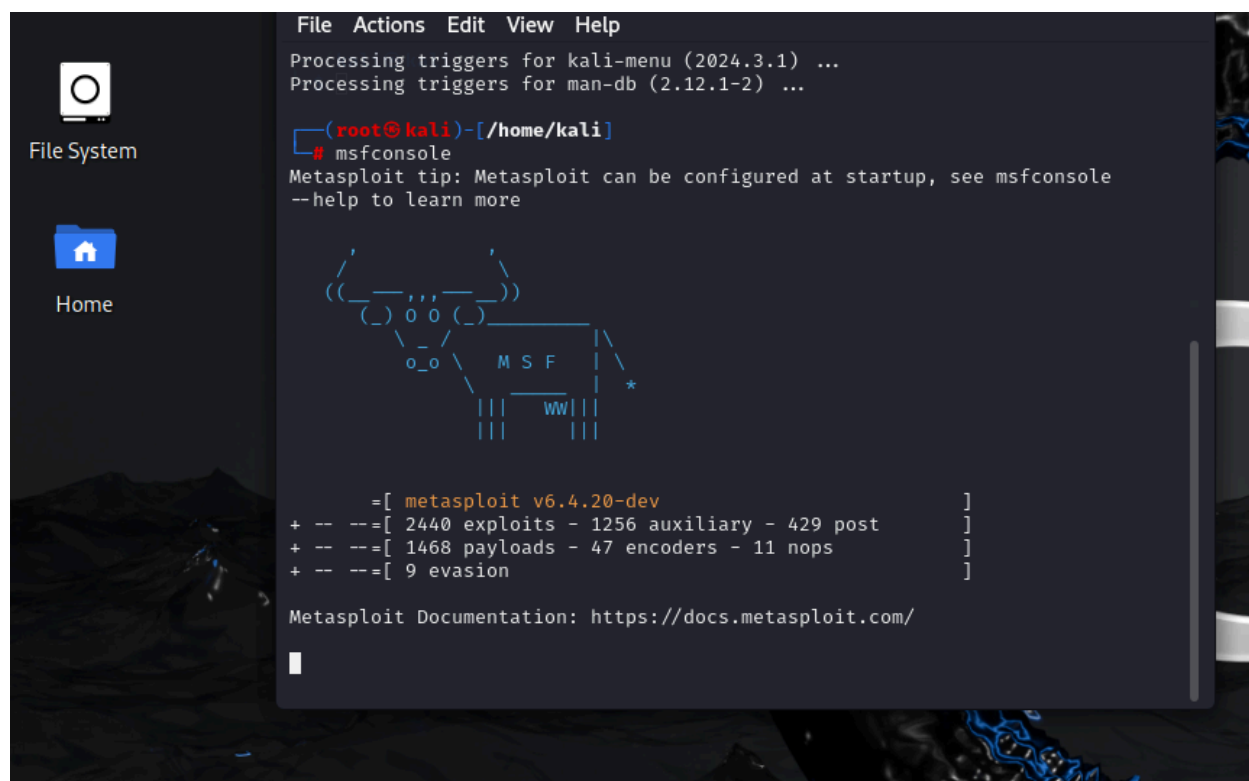


Start the **Metasploit console**:

msfconsole

This will load the Metasploit framework where you can search, use exploits, and attack your target.

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`nmap -sP <target-IP>`

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```
root@kali: /home/kali

File Actions Edit View Help

# nmap -sV 192.168.33.130 -p- --script=vuln
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-08 03:36 EDT
Nmap scan report for 192.168.33.130
Host is up (0.00032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
53/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind  2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rexecd
513/tcp   open  login    OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi  GNU Classpath grmiregistry
1524/tcp  open  bindshell Metasploitable root shell
2049/tcp  open  nfs      2-4 (RPC #100003)
2121/tcp  open  ftp      ProFTPD 1.3.1
3306/tcp  open  mysql    MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc       VNC (protocol 3.3)
6000/tcp  open  X11      (access denied)
6667/tcp  open  irc      UnrealIRCd
8009/tcp  open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp  open  http     Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:0E:73:E6 (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.07 seconds

(root@kali)-[/home/kali]
#
```

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10. Best Practices for Learning

- **Documentation:** Keep detailed notes of every action you perform.
- **Isolation:** Ensure your test lab is isolated from the internet to avoid unintended consequences.
- **Practice:** Use platforms like **TryHackMe** and **Hack The Box** to further your skills.
- **Research:** Always keep up-to-date with the latest vulnerabilities and exploits.

Resource :

01. <https://www.kali.org/>
02. <https://www.vmware.com/>
03. <https://www.virtualbox.org/>